# exam

Naro_KUOCH

2026-09-20

```
#install.packages("funFEM")
library(funFEM)
```

```
## Warning: package 'funFEM' was built under R version 4.5.3

## Loading required package: MASS

## Loading required package: fda

## Warning: package 'fda' was built under R version 4.5.3

## Loading required package: splines

## Loading required package: fds

## Warning: package 'fds' was built under R version 4.5.3

## Loading required package: rainbow

## Warning: package 'rainbow' was built under R version 4.5.3

## Loading required package: pcaPP

## Warning: package 'pcaPP' was built under R version 4.5.3

## Loading required package: RCurl

## Warning: package 'RCurl' was built under R version 4.5.3

## Loading required package: deSolve

## Warning: package 'deSolve' was built under R version 4.5.3

##
## Attaching package: 'fda'
```

```
## The following object is masked from 'package:graphics':
##
##     matplot

## The following object is masked from 'package:datasets':
##
##     gait

## Loading required package: elasticnet

## Warning: package 'elasticnet' was built under R version 4.5.2

## Loading required package: lars

## Warning: package 'lars' was built under R version 4.5.2

## Loaded lars 1.3
```

```
data(velib)
data = velib$data
dim(data)
```

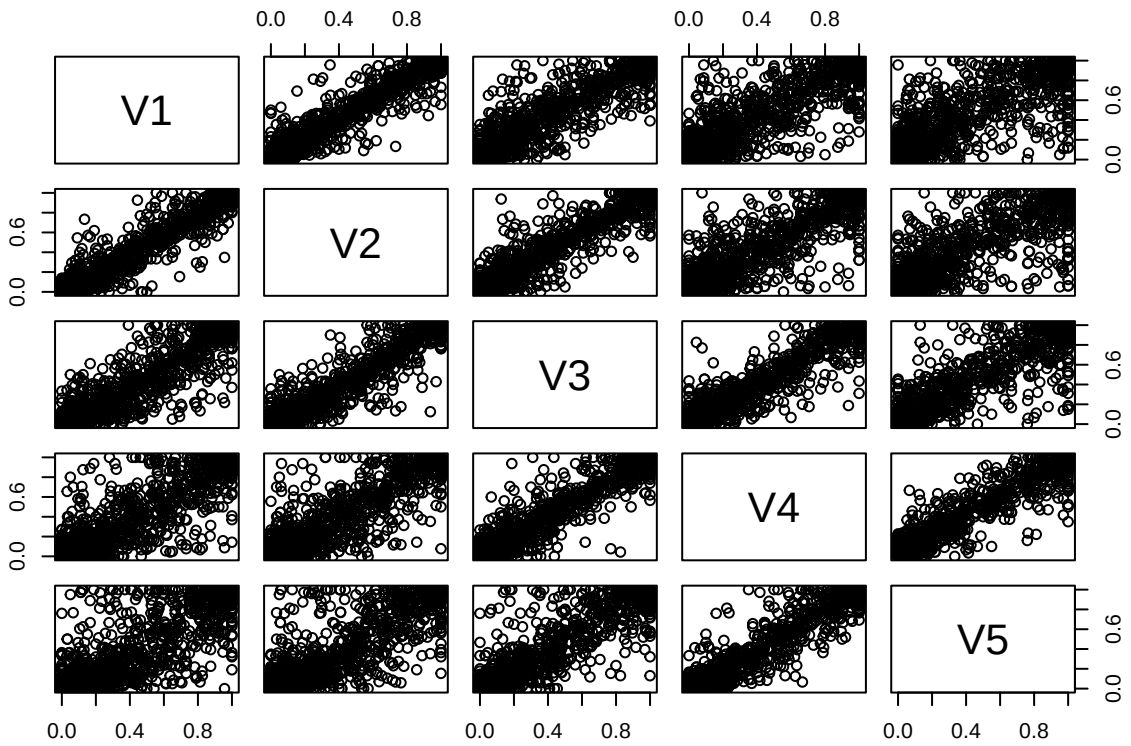
```
## [1] 1189  181
```

we have 181 variables and 1189 observations. Each observation is the loading profile of a station and each column is a time point.

```
names(data)
```

```
##   [1] "V1"   "V2"   "V3"   "V4"   "V5"   "V6"   "V7"   "V8"   "V9"   "V10"
##  [11] "V11"  "V12"  "V13"  "V14"  "V15"  "V16"  "V17"  "V18"  "V19"  "V20"
##  [21] "V21"  "V22"  "V23"  "V24"  "V25"  "V26"  "V27"  "V28"  "V29"  "V30"
##  [31] "V31"  "V32"  "V33"  "V34"  "V35"  "V36"  "V37"  "V38"  "V39"  "V40"
##  [41] "V41"  "V42"  "V43"  "V44"  "V45"  "V46"  "V47"  "V48"  "V49"  "V50"
##  [51] "V51"  "V52"  "V53"  "V54"  "V55"  "V56"  "V57"  "V58"  "V59"  "V60"
##  [61] "V61"  "V62"  "V63"  "V64"  "V65"  "V66"  "V67"  "V68"  "V69"  "V70"
##  [71] "V71"  "V72"  "V73"  "V74"  "V75"  "V76"  "V77"  "V78"  "V79"  "V80"
##  [81] "V81"  "V82"  "V83"  "V84"  "V85"  "V86"  "V87"  "V88"  "V89"  "V90"
##  [91] "V91"  "V92"  "V93"  "V94"  "V95"  "V96"  "V97"  "V98"  "V99"  "V100"
## [101] "V101" "V102" "V103" "V104" "V105" "V106" "V107" "V108" "V109" "V110"
## [111] "V111" "V112" "V113" "V114" "V115" "V116" "V117" "V118" "V119" "V120"
## [121] "V121" "V122" "V123" "V124" "V125" "V126" "V127" "V128" "V129" "V130"
## [131] "V131" "V132" "V133" "V134" "V135" "V136" "V137" "V138" "V139" "V140"
## [141] "V141" "V142" "V143" "V144" "V145" "V146" "V147" "V148" "V149" "V150"
## [151] "V151" "V152" "V153" "V154" "V155" "V156" "V157" "V158" "V159" "V160"
## [161] "V161" "V162" "V163" "V164" "V165" "V166" "V167" "V168" "V169" "V170"
## [171] "V171" "V172" "V173" "V174" "V175" "V176" "V177" "V178" "V179" "V180"
## [181] "V181"
```

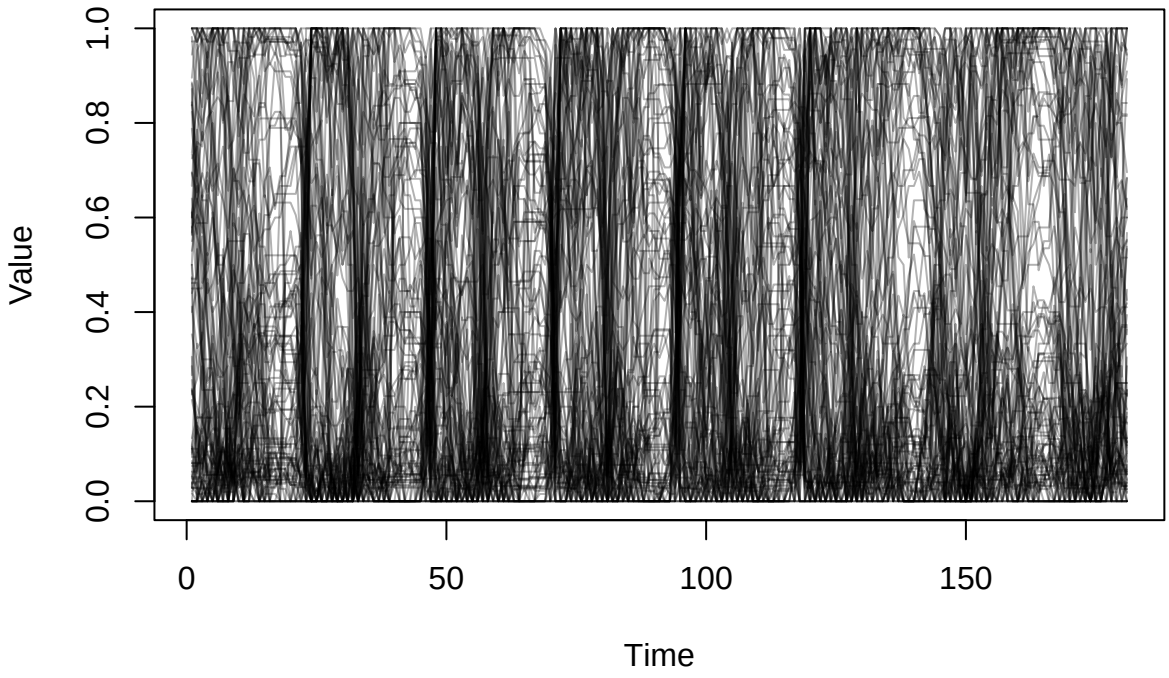
```r
pairs(data[,1:5])
```




```r
set.seed(123)

idx <- sample(nrow(data), 100)

matplot(
  t(data[idx, ]),
  type = "l",
  lty = 1,
  col = rgb(0, 0, 0, alpha = 0.3),
  xlab = "Time",
  ylab = "Value"
)
```




```
pc = princomp(data)
summary(pc)
```

```
## Importance of components:
##                           Comp.1    Comp.2     Comp.3     Comp.4     Comp.5
## Standard deviation     2.94595 2.2256993 1.02580832 0.92939356 0.87593862
## Proportion of Variance 0.40575 0.2316012 0.04919718 0.04038379 0.03587196
## Cumulative Proportion  0.40575 0.6373512 0.68654837 0.72693216 0.76280412
##                           Comp.6     Comp.7     Comp.8     Comp.9    Comp.10
## Standard deviation     0.6317361 0.60541555 0.56374901 0.51267364 0.47822454
## Proportion of Variance 0.0186586 0.01713621 0.01485865 0.01228824 0.01069231
## Cumulative Proportion  0.7814627 0.79859893 0.81345758 0.82574582 0.83643813
##                           Comp.11     Comp.12     Comp.13     Comp.14
## Standard deviation     0.46299693 0.443055795 0.406931634 0.386061822
## Proportion of Variance 0.01002222 0.009177504 0.007741955 0.006968214
## Cumulative Proportion  0.84646035 0.855637855 0.863379810 0.870348024
##                           Comp.15     Comp.16     Comp.17     Comp.18
## Standard deviation     0.377796432 0.376243427 0.354901211 0.349330399
## Proportion of Variance 0.006673036 0.006618287 0.005888744 0.005705327
## Cumulative Proportion  0.877021059 0.883639346 0.889528091 0.895233417
##                           Comp.19     Comp.20     Comp.21     Comp.22
## Standard deviation     0.330714512 0.312363821 0.290256192 0.279110479
## Proportion of Variance 0.005113453 0.004561726 0.003938862 0.003642168
## Cumulative Proportion  0.900346870 0.904908596 0.908847458 0.912489626
##                           Comp.23     Comp.24     Comp.25     Comp.26
```

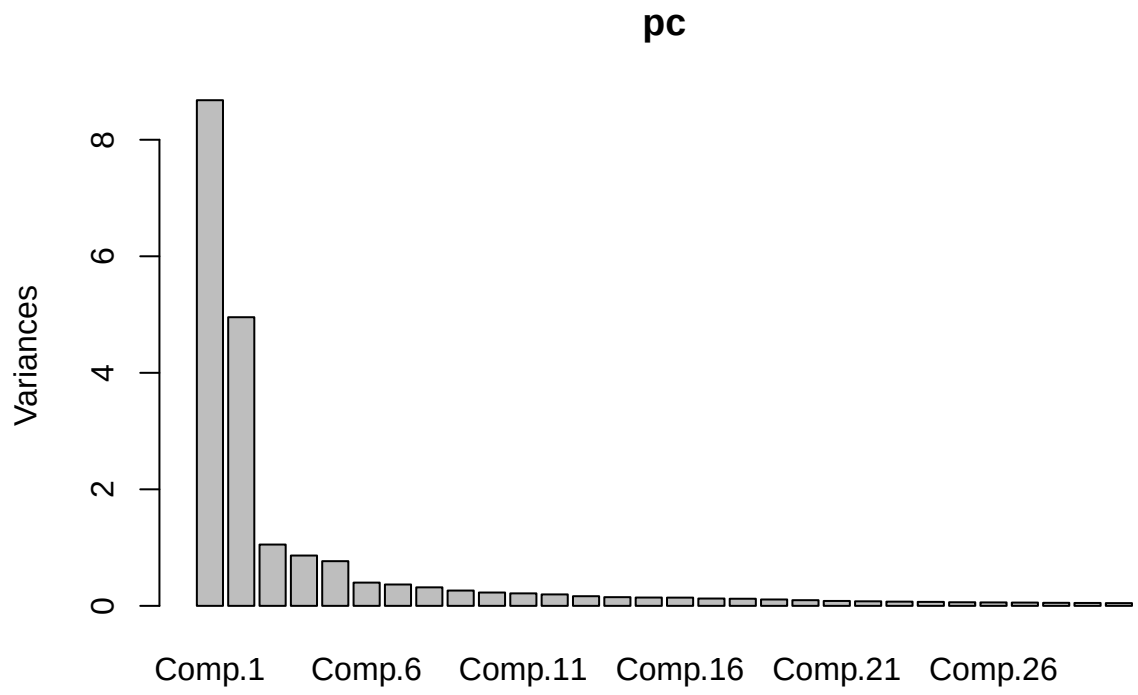
```
## Standard deviation     0.268512977 0.261121155 0.248672422 0.243156650
## Proportion of Variance 0.003370841 0.003187806 0.002891099 0.002764267
## Cumulative Proportion  0.915860467 0.919048273 0.921939372 0.924703640
##                            Comp.27     Comp.28     Comp.29     Comp.30
## Standard deviation     0.237264997 0.228302278 0.223770452 0.217401416
## Proportion of Variance 0.002631935 0.002436847 0.002341064 0.002209696
## Cumulative Proportion  0.927335574 0.929772421 0.932113485 0.934323181
##                            Comp.31     Comp.32     Comp.33     Comp.34
## Standard deviation     0.208576385 0.204578833 0.199507628 0.196939638
## Proportion of Variance 0.002033939 0.001956722 0.001860916 0.001813318
## Cumulative Proportion  0.936357120 0.938313842 0.940174758 0.941988076
##                            Comp.35     Comp.36     Comp.37     Comp.38
## Standard deviation     0.190072024 0.186770482 0.184017194 0.177732700
## Proportion of Variance 0.001689056 0.001630888 0.001583159 0.001476871
## Cumulative Proportion  0.943677133 0.945308021 0.946891180 0.948368050
##                            Comp.39     Comp.40     Comp.41     Comp.42
## Standard deviation     0.174447155 0.166038880 0.165230754 0.161548669
## Proportion of Variance 0.001422773 0.001288924 0.001276408 0.001220154
## Cumulative Proportion  0.949790823 0.951079747 0.952356155 0.953576309
##                            Comp.43    Comp.44     Comp.45     Comp.46
## Standard deviation     0.157359122 0.15542404 0.153792366 0.152869407
## Proportion of Variance 0.001157688 0.00112939 0.001105802 0.001092569
## Cumulative Proportion  0.954733997 0.95586339 0.956969189 0.958061758
##                            Comp.47     Comp.48      Comp.49      Comp.50
## Standard deviation     0.148939695 0.148570875 0.1460866605 0.1439313468
## Proportion of Variance 0.001037119 0.001031989 0.0009977664 0.0009685421
## Cumulative Proportion  0.959098877 0.960130867 0.9611286330 0.9620971751
##                             Comp.51      Comp.52      Comp.53      Comp.54
## Standard deviation     0.1390159053 0.1379226793 0.1359322259 0.1339224099
## Proportion of Variance 0.0009035178 0.0008893631 0.0008638784 0.0008385216
## Cumulative Proportion  0.9630006929 0.9638900561 0.9647539344 0.9655924560
##                             Comp.55      Comp.56      Comp.57      Comp.58
## Standard deviation     0.1313632637 0.1297716417 0.1270937156 0.1259762804
## Proportion of Variance 0.0008067809 0.0007873491 0.0007551894 0.0007419682
## Cumulative Proportion  0.9663992369 0.9671865860 0.9679417755 0.9686837437
##                             Comp.59      Comp.60      Comp.61      Comp.62
## Standard deviation     0.1238803368 0.1217550325 0.1205850195 0.1183895238
## Proportion of Variance 0.0007174845 0.0006930772 0.0006798208 0.0006552911
## Cumulative Proportion  0.9694012282 0.9700943053 0.9707741261 0.9714294173
##                            Comp.63      Comp.64      Comp.65      Comp.66
## Standard deviation     0.117380887 0.1160706994 0.1151703530 0.1127350405
## Proportion of Variance 0.000644173 0.0006298729 0.0006201391 0.0005941904
## Cumulative Proportion  0.972073590 0.9727034632 0.9733236023 0.9739177927
##                             Comp.67      Comp.68      Comp.69      Comp.70
## Standard deviation     0.1113493010 0.1103172769 0.1098202067 0.1081682408
## Proportion of Variance 0.0005796725 0.0005689771 0.0005638613 0.0005470251
## Cumulative Proportion  0.9744974652 0.9750664424 0.9756303036 0.9761773288
##                             Comp.71      Comp.72      Comp.73      Comp.74
## Standard deviation     0.1064078420 0.1055582784 0.1041190511 0.1037252808
## Proportion of Variance 0.0005293647 0.0005209456 0.0005068368 0.0005030104
## Cumulative Proportion  0.9767066935 0.9772276391 0.9777344759 0.9782374863
##                             Comp.75      Comp.76      Comp.77      Comp.78
## Standard deviation     0.1015509833 0.1008202818 0.1003858381 0.0998311357
## Proportion of Variance 0.0004821432 0.0004752297 0.0004711429 0.0004659505
```

```
## Cumulative Proportion  0.9787196294 0.9791948591 0.9796660020 0.9801319525
##                            Comp.79     Comp.80      Comp.81      Comp.82
## Standard deviation     0.0970713030 0.096681599 0.0956133567 0.0948554804
## Proportion of Variance 0.0004405442 0.000437014 0.0004274102 0.0004206613
## Cumulative Proportion  0.9805724967 0.981009511 0.9814369209 0.9818575823
##                            Comp.83      Comp.84      Comp.85      Comp.86
## Standard deviation     0.0946714286 0.0939952905 0.0924790696 0.0918238766
## Proportion of Variance 0.0004190305 0.0004130665 0.0003998477 0.0003942022
## Cumulative Proportion  0.9822766127 0.9826896792 0.9830895269 0.9834837291
##                            Comp.87     Comp.88      Comp.89      Comp.90
## Standard deviation     0.0893217774 0.089257379 0.0879328543 0.0878895166
## Proportion of Variance 0.0003730117 0.000372474 0.0003615015 0.0003611452
## Cumulative Proportion  0.9838567408 0.984229215 0.9845907163 0.9849518616
##                            Comp.91      Comp.92      Comp.93      Comp.94
## Standard deviation     0.0867809688 0.0865714520 0.0851138186 0.0848290290
## Proportion of Variance 0.0003520925 0.0003503944 0.0003386943 0.0003364316
## Cumulative Proportion  0.9853039540 0.9856543484 0.9859930427 0.9863294743
##                            Comp.95      Comp.96      Comp.97      Comp.98
## Standard deviation     0.0838186602 0.0826239908 0.0817049255 0.0809671832
## Proportion of Variance 0.0003284651 0.0003191685 0.0003121075 0.0003064967
## Cumulative Proportion  0.9866579393 0.9869771079 0.9872892154 0.9875957121
##                            Comp.99     Comp.100     Comp.101     Comp.102
## Standard deviation     0.0802433963 0.0794694127 0.0786168066 0.0780133904
## Proportion of Variance 0.0003010415 0.0002952621 0.0002889605 0.0002845418
## Cumulative Proportion  0.9878967536 0.9881920157 0.9884809763 0.9887655180
##                           Comp.103     Comp.104     Comp.105     Comp.106
## Standard deviation     0.0779162124 0.0772036247 0.0762127796 0.0760266046
## Proportion of Variance 0.0002838333 0.0002786655 0.0002715585 0.0002702333
## Cumulative Proportion  0.9890493514 0.9893280168 0.9895995753 0.9898698086
##                           Comp.107    Comp.108     Comp.109     Comp.110
## Standard deviation     0.0754600119 0.074074330 0.0737992484 0.0734649174
## Proportion of Variance 0.0002662205 0.000256533 0.0002546312 0.0002523293
## Cumulative Proportion  0.9901360291 0.990392562 0.9906471933 0.9908995226
##                           Comp.111     Comp.112     Comp.113     Comp.114
## Standard deviation     0.0729416810 0.0720596168 0.0717869767 0.0714430478
## Proportion of Variance 0.0002487478 0.0002427681 0.0002409345 0.0002386315
## Cumulative Proportion  0.9911482705 0.9913910386 0.9916319731 0.9918706046
##                           Comp.115     Comp.116     Comp.117     Comp.118
## Standard deviation     0.0703019158 0.0696973470 0.0689548743 0.0684021902
## Proportion of Variance 0.0002310692 0.0002271121 0.0002222991 0.0002187498
## Cumulative Proportion  0.9921016738 0.9923287859 0.9925510850 0.9927698348
##                           Comp.119     Comp.120     Comp.121    Comp.122
## Standard deviation     0.0678123894 0.0675109121 0.0663417012 0.065857371
## Proportion of Variance 0.0002149938 0.0002130864 0.0002057695 0.000202776
## Cumulative Proportion  0.9929848286 0.9931979149 0.9934036844 0.993606460
##                           Comp.123    Comp.124     Comp.125     Comp.126
## Standard deviation     0.0654266783 0.064769636 0.0643689521 0.0634974852
## Proportion of Variance 0.0002001324 0.000196133 0.0001937138 0.0001885041
## Cumulative Proportion  0.9938065928 0.994002726 0.9941964397 0.9943849438
##                           Comp.127     Comp.128     Comp.129     Comp.130
## Standard deviation     0.0628417159 0.0626617018 0.0618450440 0.0612420106
## Proportion of Variance 0.0001846307 0.0001835744 0.0001788206 0.0001753503
## Cumulative Proportion  0.9945695744 0.9947531488 0.9949319694 0.9951073198
##                           Comp.131     Comp.132     Comp.133     Comp.134
```

```
## Standard deviation     0.0610495180 0.0602726726 0.0598603275 0.0590165988
## Proportion of Variance 0.0001742498 0.0001698434 0.0001675274 0.0001628381
## Cumulative Proportion  0.9952815695 0.9954514129 0.9956189403 0.9957817785
##                           Comp.135     Comp.136     Comp.137     Comp.138
## Standard deviation     0.058959282 0.0582037042 0.0575064364 0.0571410692
## Proportion of Variance 0.000162522 0.0001583832 0.0001546111 0.0001526527
## Cumulative Proportion  0.995944300 0.9961026836 0.9962572947 0.9964099474
##                           Comp.139     Comp.140     Comp.141     Comp.142
## Standard deviation     0.0566058710 0.0561097703 0.0559422172 0.0552800844
## Proportion of Variance 0.0001498065 0.0001471922 0.0001463144 0.0001428713
## Cumulative Proportion  0.9965597539 0.9967069461 0.9968532605 0.9969961319
##                           Comp.143    Comp.144     Comp.145     Comp.146
## Standard deviation     0.054814515 0.054325369 0.0536547122 0.0521486754
## Proportion of Variance 0.000140475 0.000137979 0.0001345933 0.0001271435
## Cumulative Proportion  0.997136607 0.997274586 0.9974091792 0.9975363227
##                           Comp.147     Comp.148     Comp.149     Comp.150
## Standard deviation     0.0519291619 0.0512557251 0.0509145857 0.0507118008
## Proportion of Variance 0.0001260754 0.0001228266 0.0001211971 0.0001202336
## Cumulative Proportion  0.9976623981 0.9977852247 0.9979064218 0.9980266554
##                           Comp.151     Comp.152     Comp.153     Comp.154
## Standard deviation     0.0497063618 0.0488269116 0.0478752944 0.0472645515
## Proportion of Variance 0.0001155132 0.0001114619 0.0001071595 0.0001044429
## Cumulative Proportion  0.9981421686 0.9982536305 0.9983607900 0.9984652328
##                           Comp.155     Comp.156     Comp.157     Comp.158
## Standard deviation     0.0464289313 0.0458769984 4.520495e-02 0.0445911931
## Proportion of Variance 0.0001007825 0.0000984006 9.553877e-05 0.0000929621
## Cumulative Proportion  0.9985660153 0.9986644159 9.987600e-01 0.9988529168
##                           Comp.159     Comp.160     Comp.161     Comp.162
## Standard deviation     4.395960e-02 4.316287e-02 4.233672e-02 4.147020e-02
## Proportion of Variance 9.034731e-05 8.710206e-05 8.379962e-05 8.040445e-05
## Cumulative Proportion  9.989433e-01 9.990304e-01 9.991142e-01 9.991946e-01
##                           Comp.163     Comp.164     Comp.165     Comp.166
## Standard deviation     3.994379e-02 3.890809e-02 3.840105e-02 3.740219e-02
## Proportion of Variance 7.459441e-05 7.077624e-05 6.894359e-05 6.540361e-05
## Cumulative Proportion  9.992692e-01 9.993399e-01 9.994089e-01 9.994743e-01
##                           Comp.167     Comp.168     Comp.169     Comp.170
## Standard deviation     0.0365687504 3.568198e-02 3.367672e-02 3.204147e-02
## Proportion of Variance 0.0000625213 5.952586e-05 5.302336e-05 4.799904e-05
## Cumulative Proportion  0.9995368094 9.995963e-01 9.996494e-01 9.996974e-01
##                           Comp.171     Comp.172     Comp.173     Comp.174
## Standard deviation     0.0306268783 2.973992e-02 2.915757e-02 2.768987e-02
## Proportion of Variance 0.0000438544 4.135113e-05 3.974754e-05 3.584674e-05
## Cumulative Proportion  0.9997412121 9.997826e-01 9.998223e-01 9.998582e-01
##                           Comp.175     Comp.176     Comp.177     Comp.178
## Standard deviation     2.443894e-02 0.0227587858 2.214726e-02 2.058651e-02
## Proportion of Variance 2.792368e-05 0.0000242162 2.293232e-05 1.981404e-05
## Cumulative Proportion  9.998861e-01 0.9999102974 9.999332e-01 9.999530e-01
##                           Comp.179     Comp.180     Comp.181
## Standard deviation     1.907157e-02 1.813142e-02 1.766009e-02
## Proportion of Variance 1.700516e-05 1.536991e-05 1.458121e-05
## Cumulative Proportion  9.999700e-01 9.999854e-01 1.000000e+00
```

If we apply the rule of the 90%, with d = 20 we already have about 90% of the full variance.

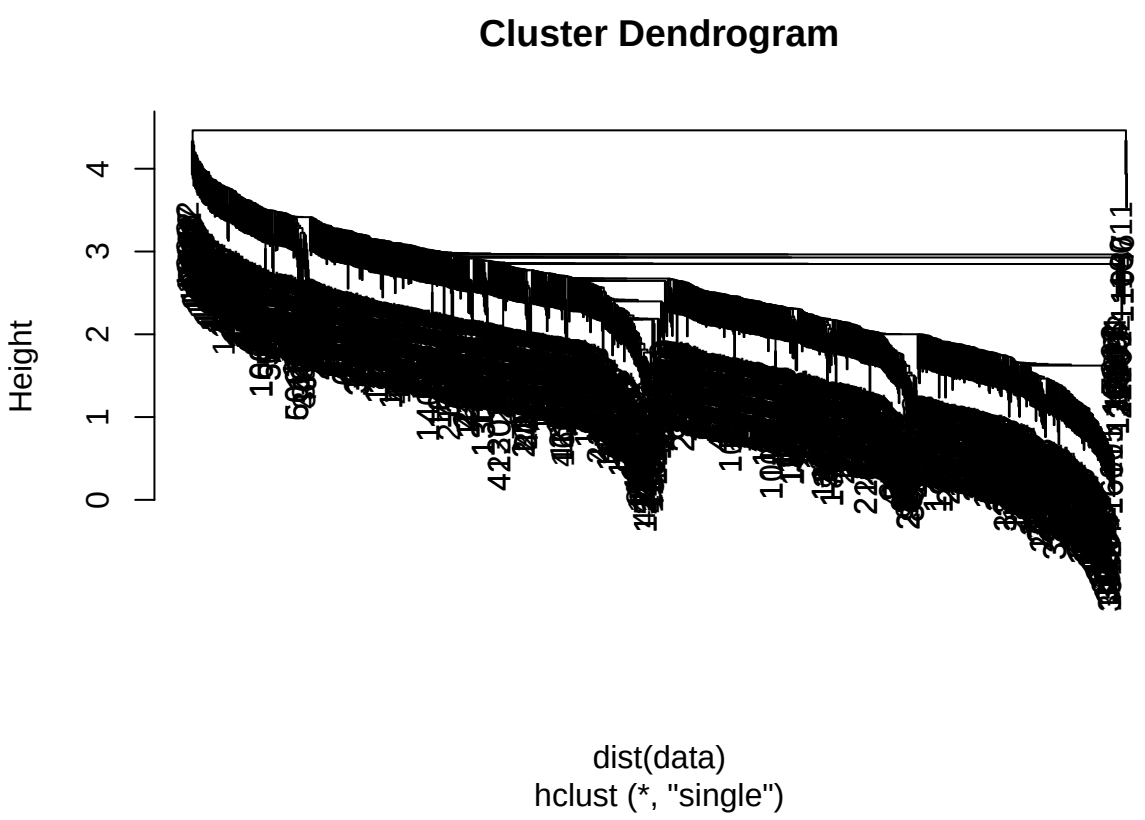
```
screeplot(pc,npcs=30)
```



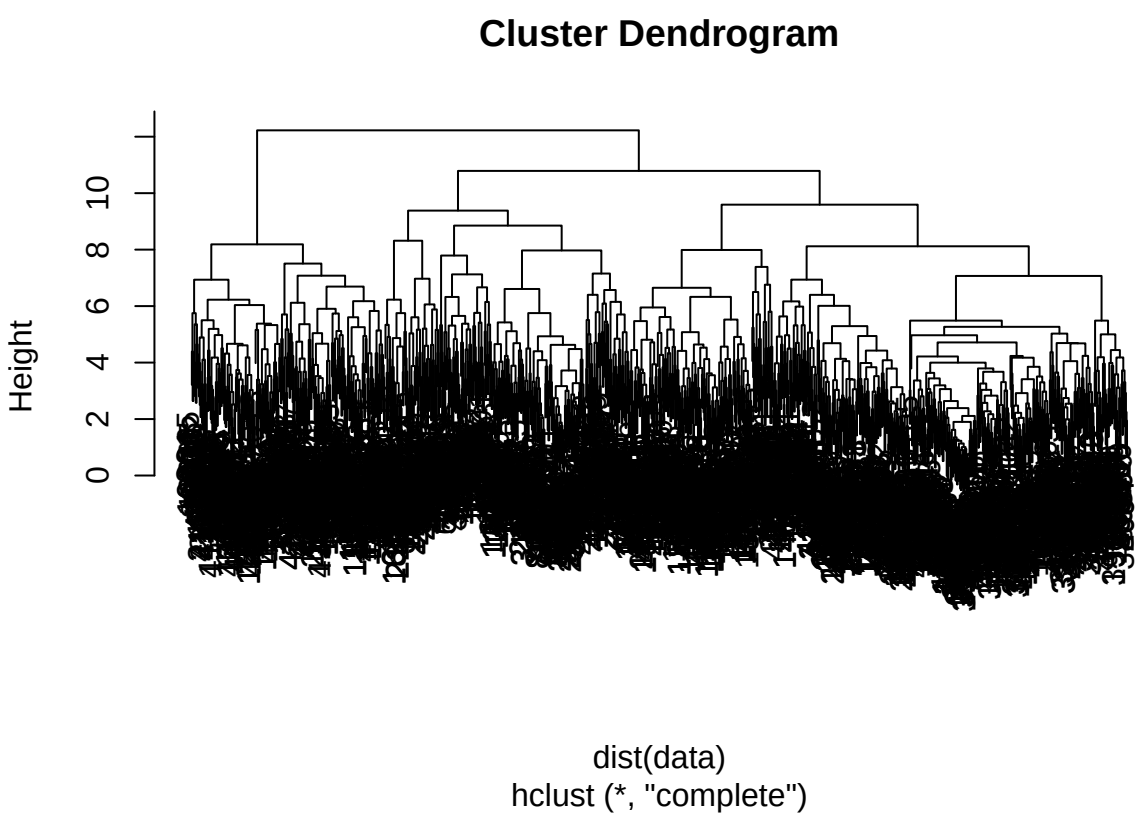

if we apply the elbow rule we can use just the first 5 components.

## Hierarchical Clustering

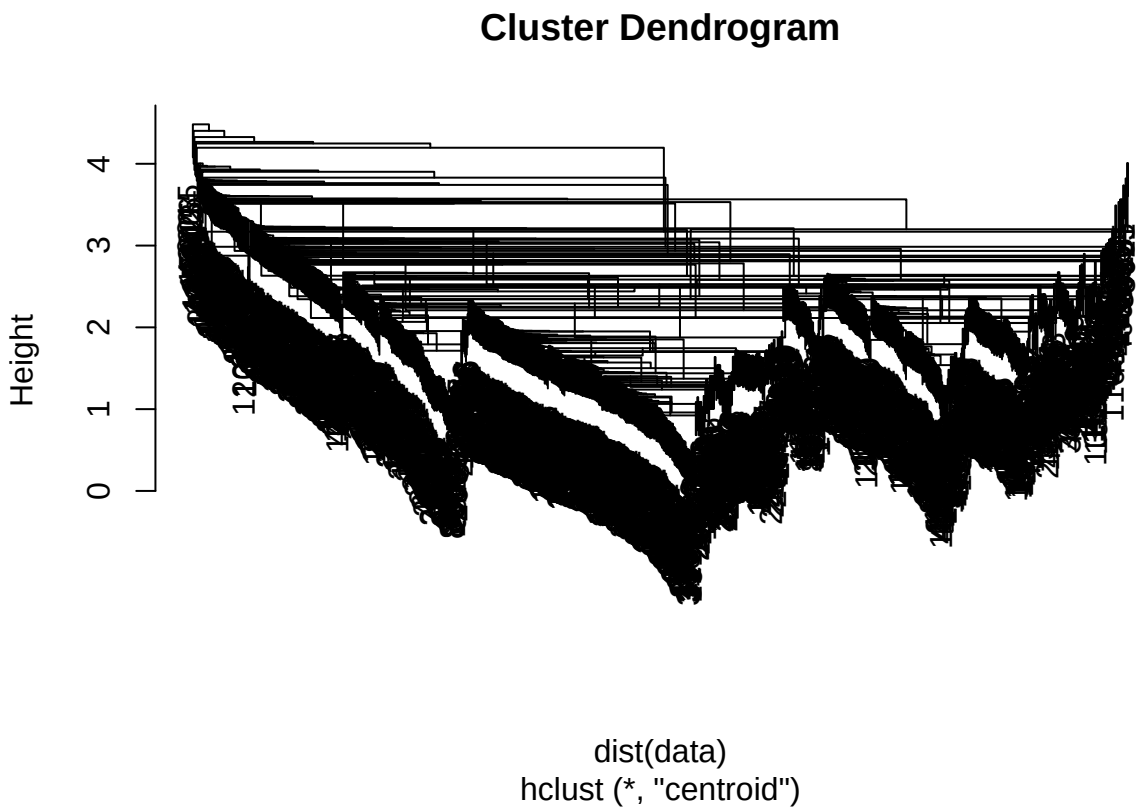
```
plot(hclust(dist(data),method = "single"))
```



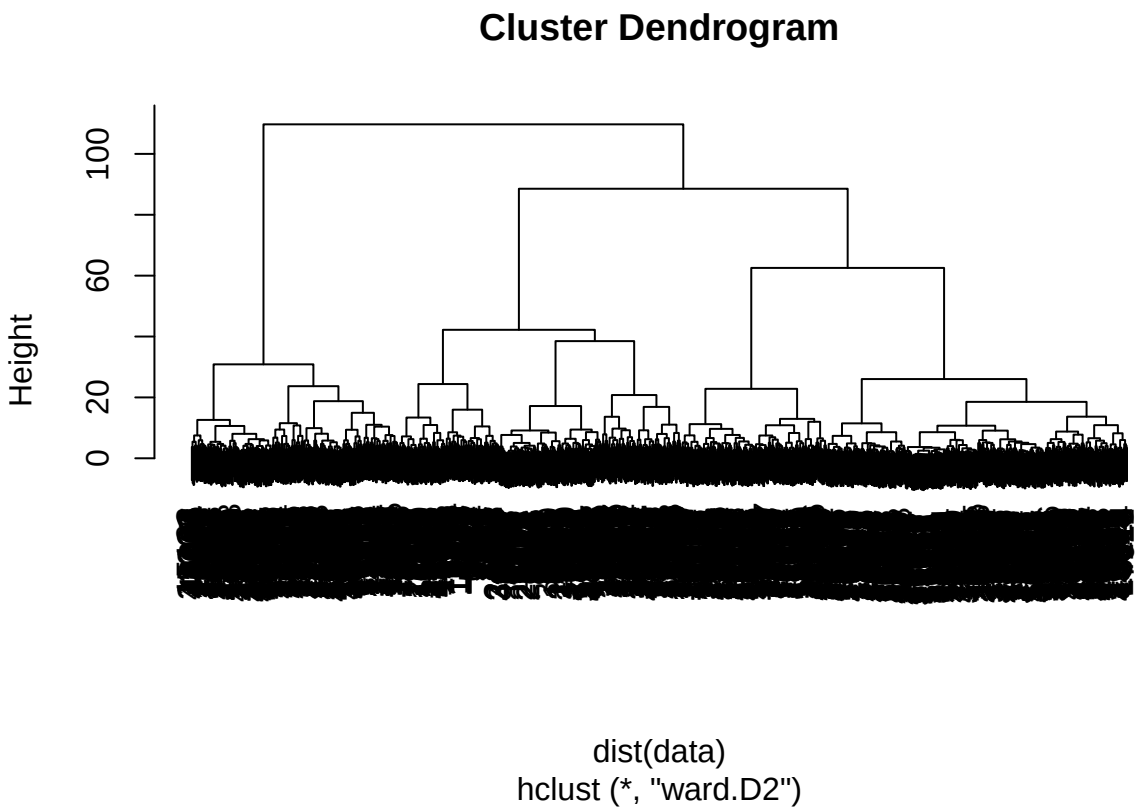

```
plot(hclust(dist(data),method = "complete"))
```




```
plot(hclust(dist(data),method = "centroid"))
```

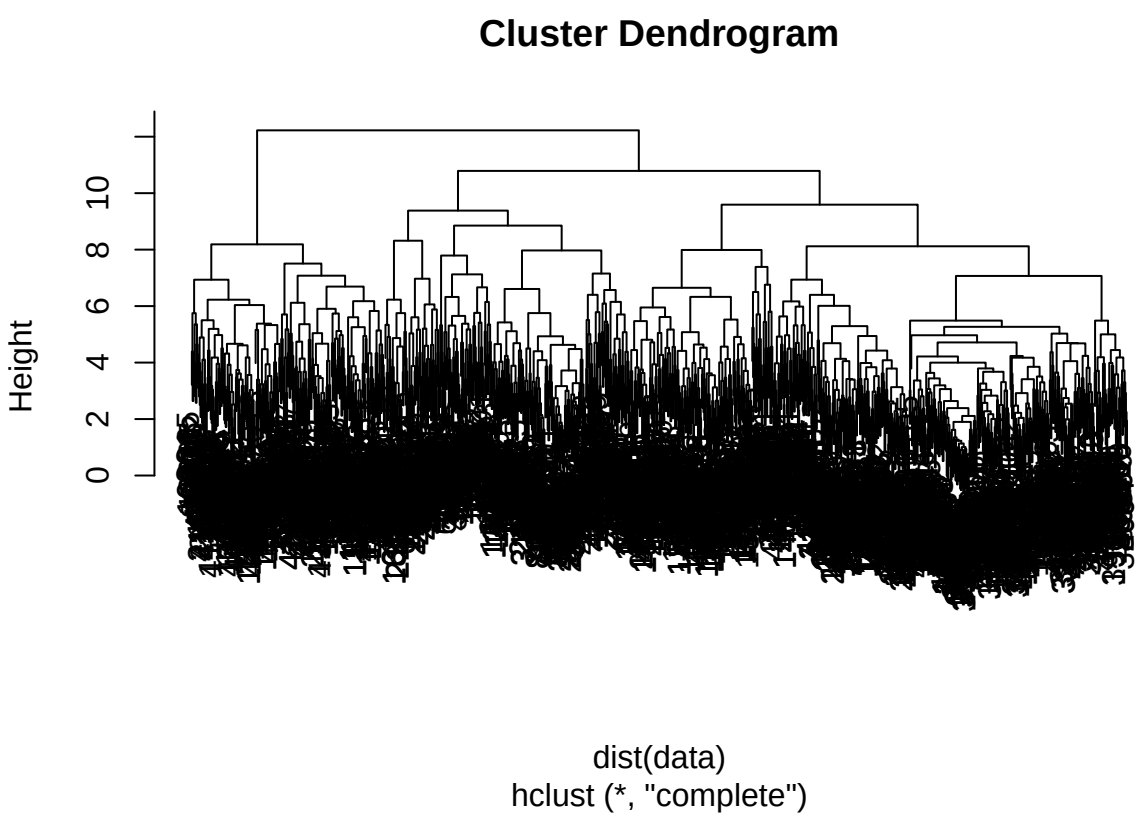



```
plot(hclust(dist(data),method = "ward.D2"))
```



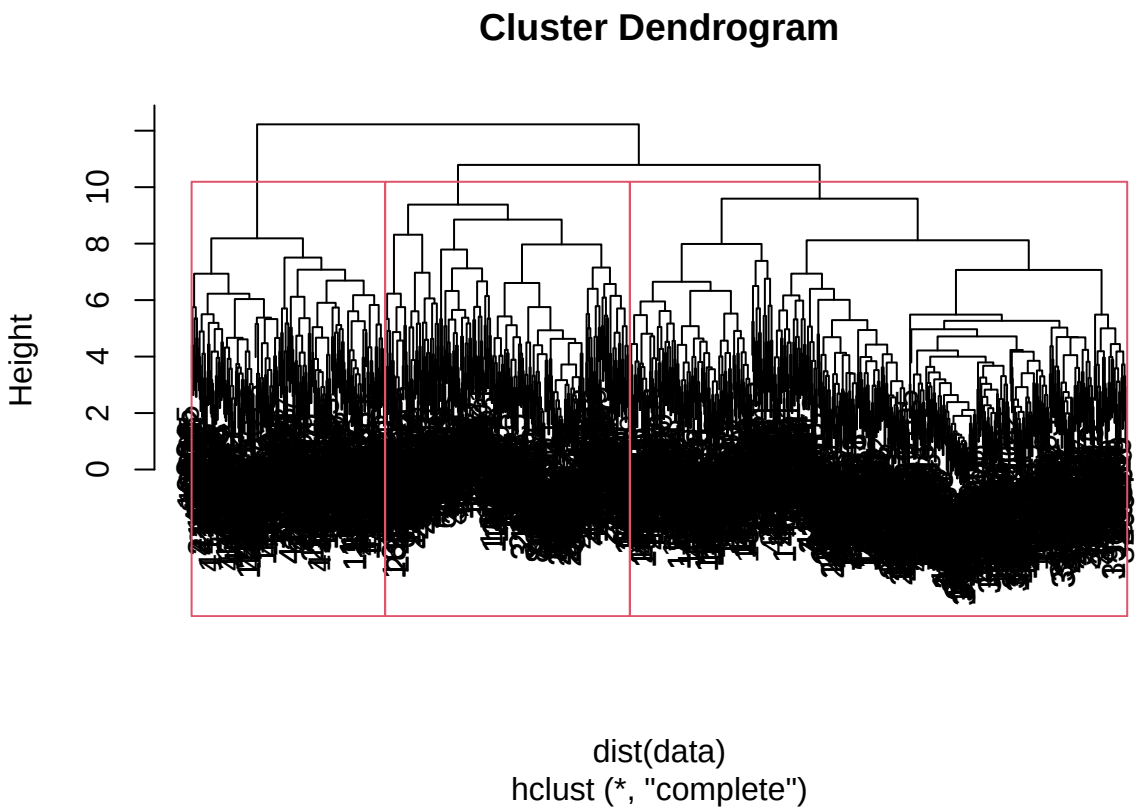

we can chose complete linkage or ward distance because the tree seems balance.

```
#complete
hc = hclust(dist(data),method = "complete")
plot(hc)
```

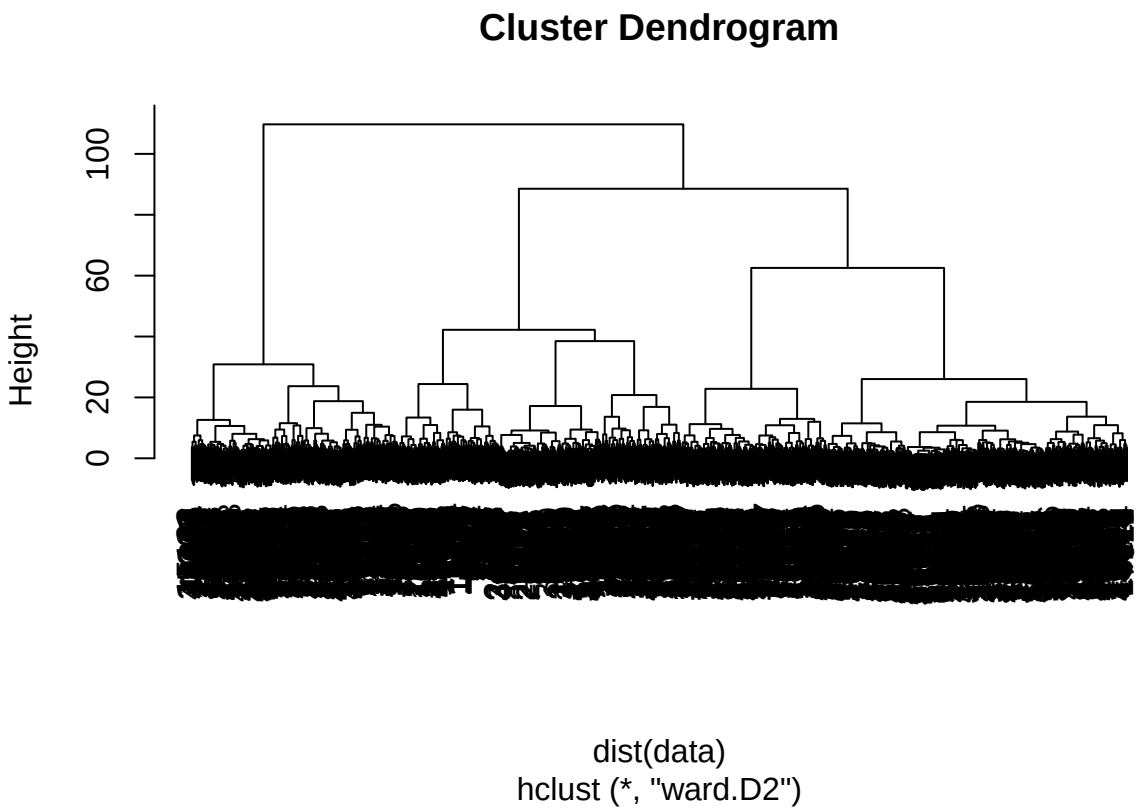



we need to cut the tree at branch that is the largest jumpe which k =3

```
plot(hc)
rect.hclust(hc,k = 3)
```

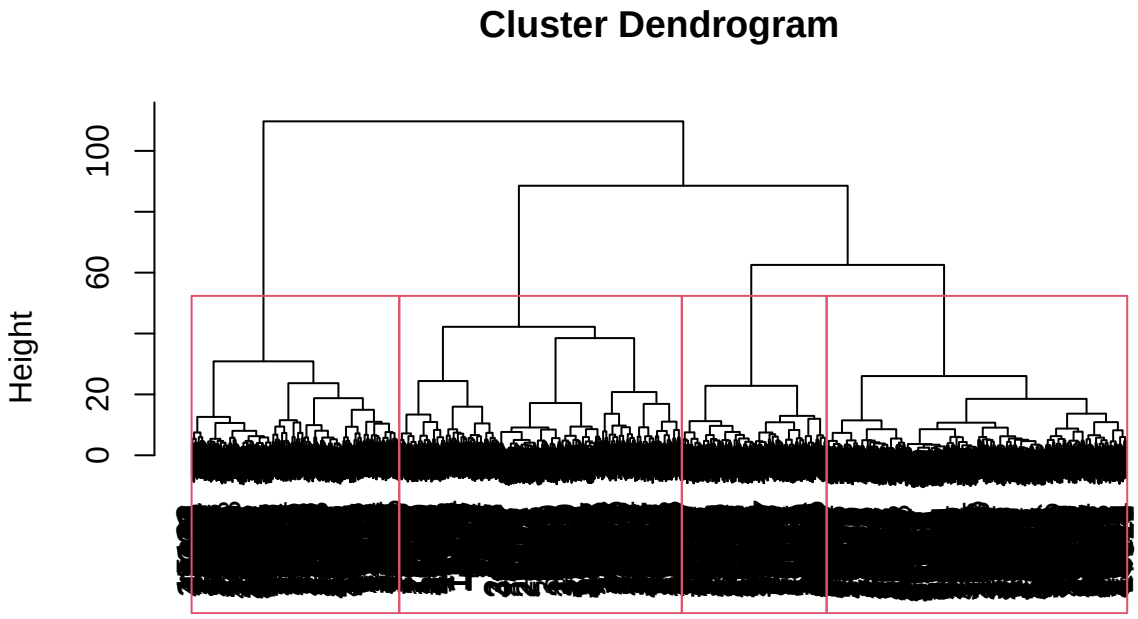



```
w_hc = hclust(dist(data),method = "ward.D2")
plot(w_hc)
```




we choose k = 4 for ward distance

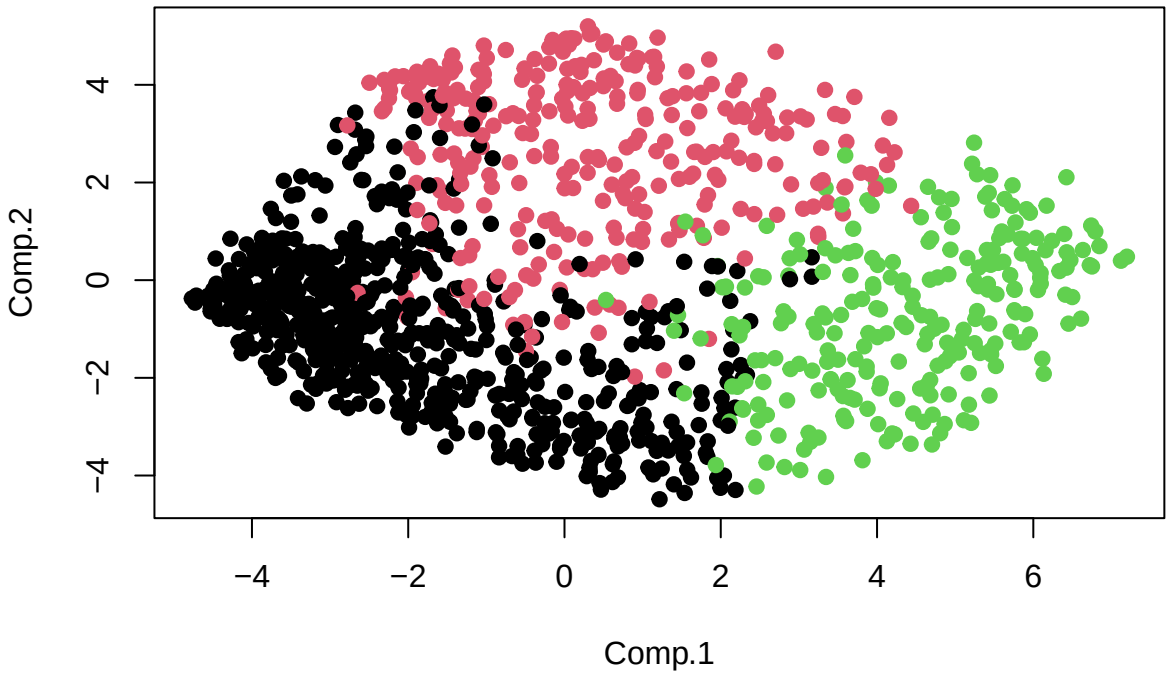
```
plot(w_hc)
rect.hclust(w_hc,k = 4)
```



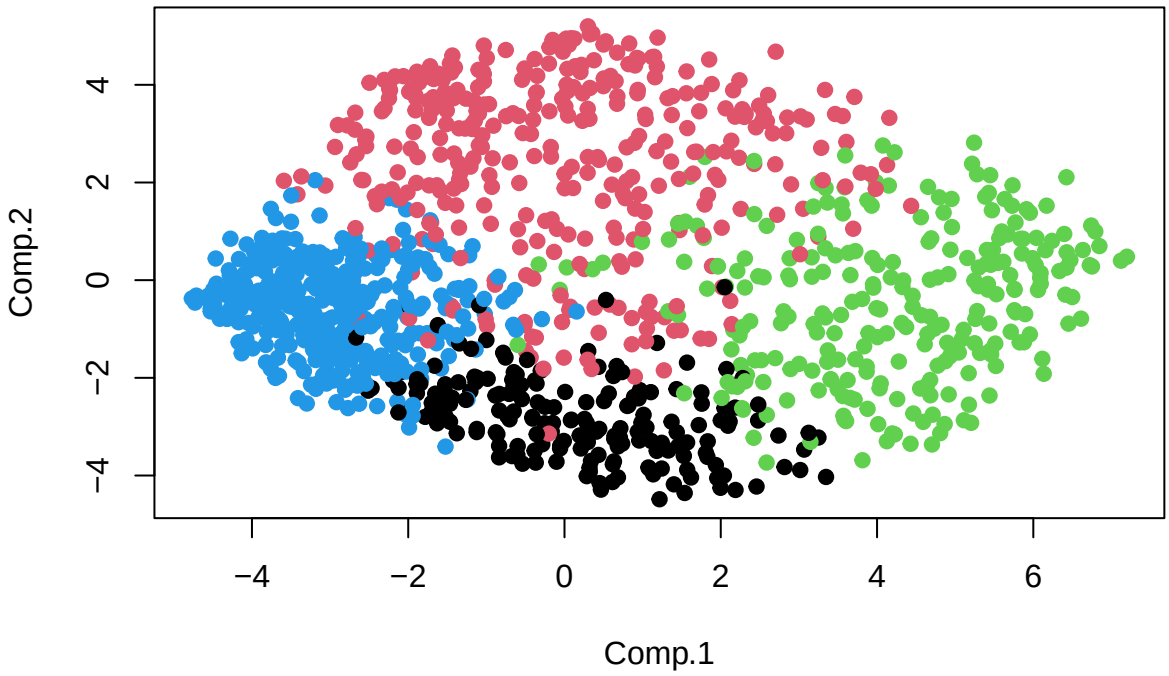

```r
X = as.matrix(data,ncol = ncol(data))

# perform PCA, then project the data on the new PCA system, take the first 2 principle components
X.2d = predict(princomp(X))[,1:2]

hc = hclust(dist(data),method = "complete")
clust = cutree(hc,k = 3)
#plot the cluster with the compete method result on the 2 pca components
plot(X.2d,col=clust,pch=19)
```




```
w_hc = hclust(dist(data),method = "ward.D2")
w_clust = cutree(w_hc,k =4 )
#plot the cluster with the compete method result on the 2 pca components
plot(X.2d,col=w_clust,pch=19)
```




## K-means

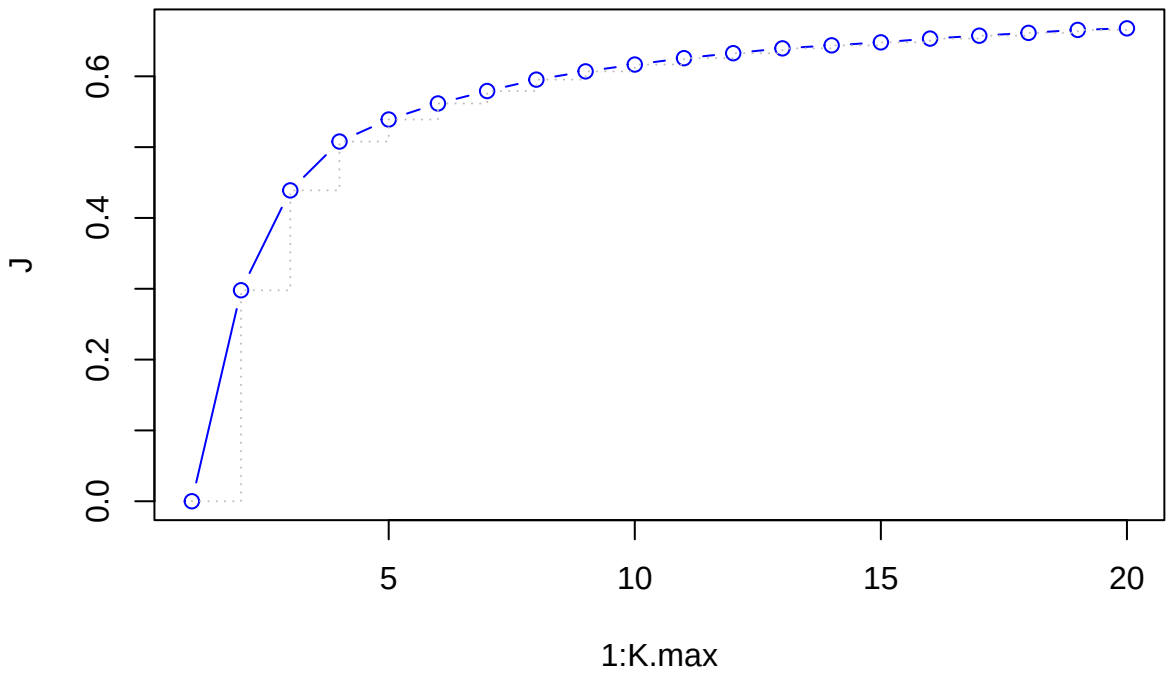
find the best k for kmeans using criterions

$$J(K) = \frac{B(K)}{S}$$

```
K.max = 20
J = rep(NA,K.max)
for (k in 1:K.max){
  out = kmeans(data, centers = k, nstart = 10)
  J[k] = out$betweenss / out$totss
}
```

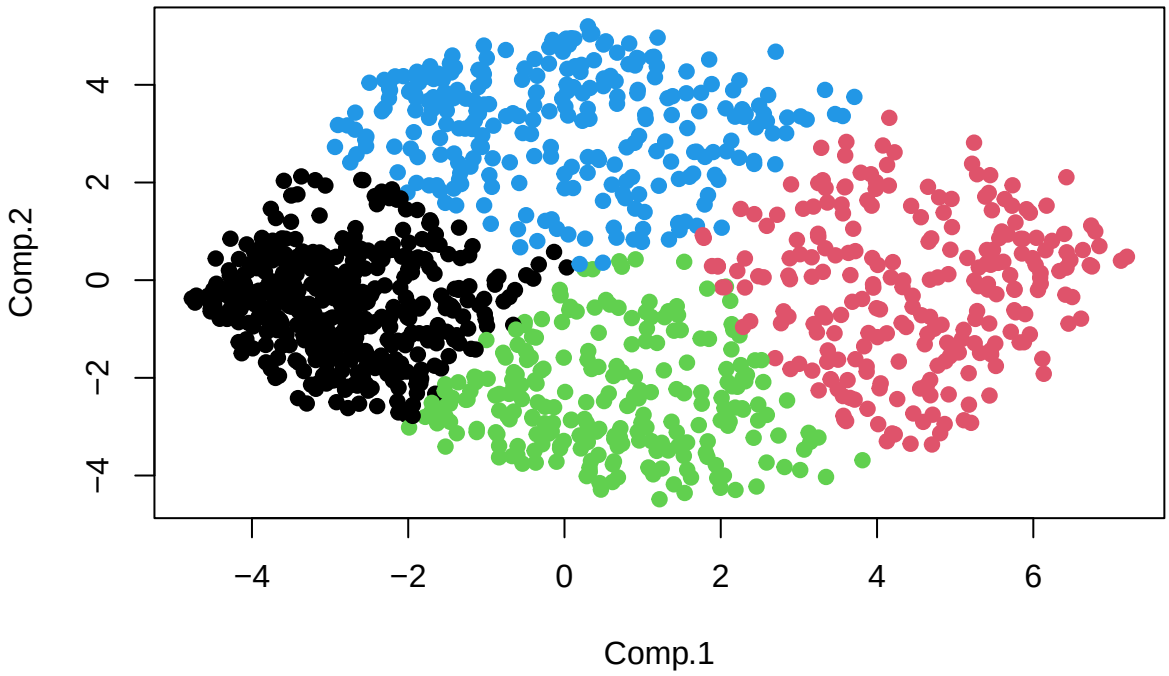
```
## Warning: did not converge in 10 iterations
```

```
plot(1:K.max,J,type='b',col='blue')
lines(1:K.max,J,type='s',lty=3,col='gray')
```



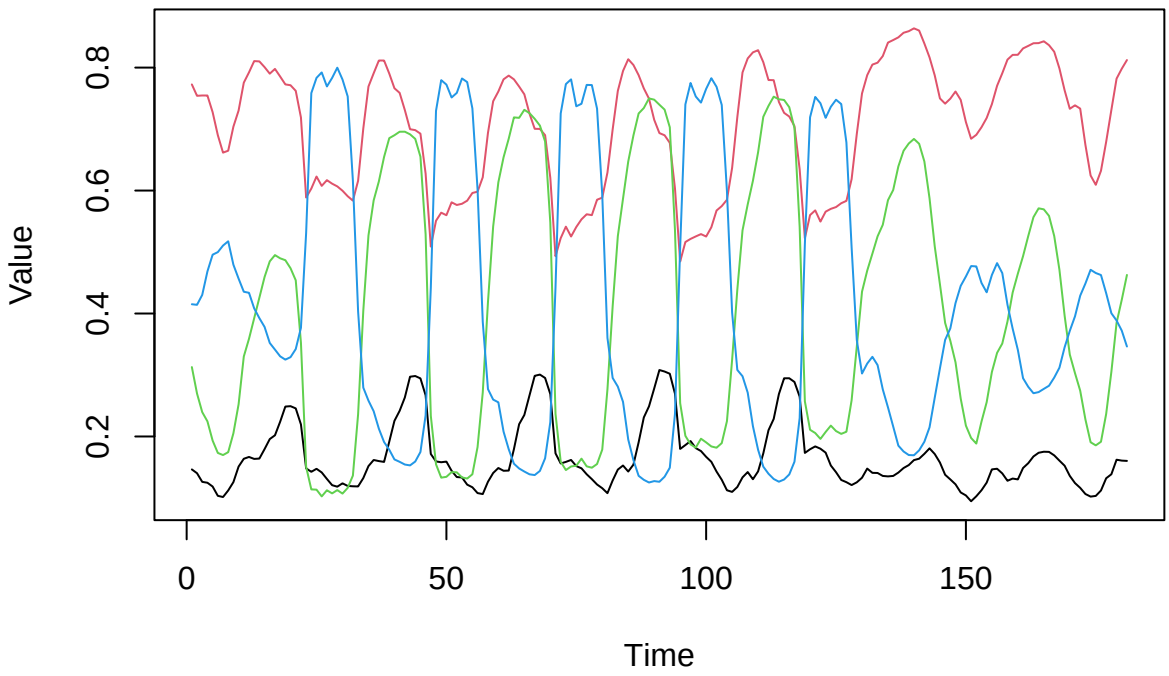

from this plot we can take K = 4 or 5. in order to compare with the previous reslut i take k=4

```
km = kmeans(data, centers = 4, nstart = 10)
plot(X.2d,col=km$cluster,pch=19)
```




From the result projected on the first 2 component of PCA, it seems that K-Mean can separate the data better.

```
matplot(
  t(km$centers),
  type = "l",
  lty = 1,
  xlab = "Time",
  ylab = "Value"
)
```




Among all the 1189 stations we can cluster them into 4 groups that have similar behavior